

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper No. STB-201: Descriptive Statistics and Distribution Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Define the coefficient of correlation. Show that it is unaffected by a change of origin but may be affected by a change of scale. [#1]
- (b) Derive the formula for the rank correlation coefficient. If (x_i, y_i) denotes the rank of the i -th pair ($i = 1, 2, \dots, n$) with $x_i - y_i = d_i$ and $x_i + y_i = n + 1$, then show that the rank correlation between X and Y is -1 . [#1]

Question 2.

- (a) Using the principle of least squares, obtain expressions for the intercept and slope of the regression line $Y = a + bX$. The regression equation between income (X) and expenditure (Y) is $Y = 500 + 50X$. If the ratio of the standard deviation of expenditure to the standard deviation of income is 60, find the coefficient of correlation between income and expenditure. [#1]
- (b) Derive the expression to measure the acute angle between two regression lines. For a certain dataset, $Y - 1.2X = 0$ and $X - 0.6Y = 0$ are the regression lines. If θ is the acute angle, find the value of $\tan \theta$. [#1]

Question 3. Explain the concept of multiple and partial correlations. In the case of three variables X_1, X_2 , and X_3 , show that:

$$1 - R_{1.23}^2 = \frac{|R|}{|R_{11}|}$$

where $|R|$ is the determinant of the correlation matrix of variables X_1, X_2 , and X_3 , and $|R_{11}|$ is the cofactor of r_{11} in $|R|$. Justify that the range of $R_{1.23}$ is between zero and one. [#1]

Question 4.

- (a) Three fair coins are tossed. Let X denote the number of heads on the first two coins and Y denote the number of tails on the last two coins. Find the joint distribution of X and Y . What will be the value of $\text{Cov}(X, Y)$? [#1]

- (b) Define the moment generating function (MGF) of a random variable and write any three of its properties. For a random variable X whose probability density function is:

$$f(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty$$

find the moment generating function of X and hence find the mean and variance of $Z = 4X - 3$. [#1]

Question 5.

- (a) Let the joint probability mass function of random variables (X, Y) be:

$$P(X = x, Y = y) = \frac{x + 2y}{18}, \quad x = 1, 2; \quad y = 1, 2$$

and zero elsewhere. Find:

- (i) $E(X)$
- (ii) $E(Y|X)$
- (iii) $E[\max(X, Y)]$

[#1]

- (b) Define Bernoulli distribution. Let $X_1, X_2, \dots, X_n$ be n independent Bernoulli random variables, each with parameter p . If $Y = \sum_{i=1}^n X_i$, obtain the probability distribution of Y . [#1]
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Question 6.

- (a) Show that the mean, median, and mode for a normal distribution are always the same. [#1]
- (b) The examination score (X) of 1000 students is normal with mean 50 and standard deviation 10. Approximately how many students will fail if the passing mark is 30? [#1]
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Question 7.

- (a) Define Gamma distribution and find its mean, harmonic mean, and standard deviation. [#1]
- (b) State and prove Chebyshev's inequality. If the mean of a random variable X is zero and $P(|X| < 2) = 1/2$, find the lower bound for the variance of X . [#1]
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Question 8.

(a) Define the negative binomial distribution and find its mean. If the probability that a randomly selected person is found honest in interrogation is $\frac{1}{1000}$, how many persons are expected to be interrogated to get the first honest person? (Provide your logic for the answer). [#1]

(b) Write short notes on any four of the following in brief:

- (i) Attributes and their analysis
- (ii) Coefficient of determination
- (iii) Bivariate normal distribution
- (iv) Intraclass correlation
- (v) Poisson distribution
- (vi) Properties of normal distribution

[#1]